# LAI Hao Ran

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## Education

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|---|---|
| 2015–2018 | Ph.D., Ecology and Evolutionary Biology, National University of Singapore. Advisors: Michiel van Breugel & Hugh Tan Tiang Wah. Thesis: *Functional ecology of tropical secondary forests*. |
| 2009–2012 | B.Sc. (Hons.) Ecology, The University of Queensland. Advisor: Margaret M. Mayfield. Thesis: *Functional recovery following logging in subtropical forests*. |

## Career summary

I am a quantitative ecologist and data scientist with over 10 years of research and teaching experience in spatial modelling and vegetation mapping. I have taught **spatial ecology** and biostatistics, twice won the **Best Teaching Assistant Award** (NUS), and supervised 6 PhD, 4 MSc and 8 undergraduate students. I have 13 peer-reviewed publications in Q1 journals as first or corresponding author. My 37 publications have been collectively cited 1,348 times with an h-index of 15. My projects have been funded by the ForestGEO Research Award (USD 10,000) and the Marsden Fast Start Fund (NZD 360,000).

## Employment

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| 2024–present | Ecologist–Data Scientist, South East Asia Rainforest Research Partnership (SEARRP), Malaysia (joint appointment by Imperial College London, UK)<br>— Virtual Ecosystem (digital twin)<br>— Spatial validation of process-based ecological models in Borneo |
| 2019–2025 | Research Fellow, School of Biological Sciences, University of Canterbury, New Zealand<br>— Species distribution and spatial modelling |
| 2019 | Visiting Research Fellow, Centre for Urban Greenery and Ecology, National Parks Board, Singapore<br>— Spatial patterns of tree diameter growth in urban streetscapes |
| 2018–2019 | Research assistant, Department of Biological Sciences, National University of Singapore, Singapore<br>— Freshwater swamp forest survey and mapping |
| 2015–2018 | Teaching assistant, National University of Singapore, Singapore<br>— *Please see Teaching section for taught courses.* |
| 2015 | Biology Olympiad trainer, National Junior College, Singapore |
| 2012–2014 | Research officer, The University of Queensland, Australia<br>— Mapping of endangered plant species |
| 2012–2014 | Course Tutor, The University of Queensland, Australia<br>— *Please see Teaching section for taught courses.* |

## Teaching

| | |
|---|---|
| 2025 | Bayesian statistics for ecologists |
| 2015–2018 | Field Studies in Biodiversity, Ecology of Terrestrial Environments, Tropical Horticulture, Quantitative Methods for Ecological Research, Comparative Botany, Ecology and the Environment, Biodiversity |
| 2012–2014 | Biostatistics, Ecology, Plant Biology |

## Supervision & Mentoring

| | |
|---|---|
| 2026 | Pin Jia Chan (PhD), Rose Foster-Dyer (PhD) |
| 2025 | Daniel Hernández (PhD), Felicia Wei Shan Leong (PhD), Tim Logan (MSc.) |
| 2024 | Sophie O'Brien (PhD), Gabrielle Koerich (PhD), Jack Anderson (MSc.) |
| 2023 | Laura van Ginkel (MSc.), Rory Lennox (MSc.) |
| 2017 | Zhong Yu Chiam (BSc. Hons) |
| 2016 | Melissa Wong Qi Yun (BSc. Hons), Amanda Ng Wei Ling (BSc. Hons), Pin Jia Chan (Undergraduate programme), Zhong Yu Chiam (Undergraduate programme), Rens Gerard Brouwer (Undergraduate programme), Srishti Arora (Undergraduate programme) |
| 2015 | Germaine Tan Su Yin (BSc. Hons) |

## Publications

∗ *denotes spatial- and mapping-related publications.*

### *Peer-reviewed articles*

Lai, H. R., Chong, K. Y., Neo, L., Wardle, D. A., Vesk, P. A. (2026). Asymmetric transferability of trait–environment relationships between native and exotic plant species in Singaporean forests differing in land-use history. *Oikos*, 1–11.

∗ Ooi, M. Y., Ho, Y. Y., Lai, H. R., Yee, A. T. K., Chong, K. Y. (2026). Forest dynamics during the thinning phase of succession following a severe windstorm in a hurricane-free tropical forest. *Journal of Tropical Ecology*, 42.

Bufford, J. L., Brandt, A. J., Sprague, R., Buddenhagen, C. E., Duncan, R. P., Harcourt, N., Hulme, P. E., Lai, H. R., McAlpine, K. G., McGrannachan, C., Thomson, F. J., Uitregt, V. van, Peltzer, D. A. (2026). Mechanisms and impacts of future plant invasions: research agendas and management needs under disruptive social-ecological change scenarios. *Oikos*, 1–17.

∗ Koerich, G., Lai, H. R., Lee, C. K., Morgan, F. J., Tonkin, J. D. (2026). Proximity to seabird colonies and water availability shape moss distributions in Antarctica. *Ecography*, 2026(1), 1–14.

Hernández-Carrasco, D., Gillis, A. J., Lai, H. R., Siqueira, T., Tonkin, J. D. (2026). Accounting for the Influence of Community Turnover Along Environmental Gradients on Compositional Uniqueness. *Ecology Letters*, 29(2), 1–12.

Lai, H. R., Burcham, D. C., Wang, J. W., Stouffer, D. B., Qiu, D. W., Yee, A. T. K. (2025). Quantifying life-history trade-offs in diameter growth for tropical tree species from a large urban inventory dataset. *Urban Forestry and Urban Greening*, 112, 128982.

∗ O'Brien, S. A., Tylianakis, J. M., Anderson, D. P., Boesing, A. L., Lai, H. R., Le Provost, G., Manning, P., Neyret, M., Blüthgen, N., Jung, K., Magdon, P., Müller, S., Schenk, N. V., Scherer-Lorenzen, M., Lavorel, S. (2025). Mobile species' responses to surrounding land use generate trade-offs and synergies among nature's contributions to people. *Proceedings of the National Academy of Sciences of the United States of America*, 122(45).

Lai, H. R. (2025). Model-based ordination for phenological studies: From controlling sampling bias to inferring temporal associations. *Methods in Ecology and Evolution*, 16, 1542–1546.

Chan, P. J., Lai, H. R., Burns, B. R., Schwendenmann, L. (2025). Potential effects of an emerging disease on tree communities around a susceptible foundation tree species. *Austral Ecology*, 50, e70102.

∗ Herbert, S. M., Tomscha, S. A., Lai, H. R., Benavidez, R., Balkwill, C. G., Ruston, P. R., Jackson, B., Deslippe, J. R. (2025). Identifying potentially suitable and accessible refugia to mitigate impacts of an emerging disease on a rare tree. *Conservation Biology*, e70088. https://doi.org/10.1111/cobi.70088

Dehling, D. M., Lai, H. R., Stouffer, D. B. (2025). Eltonian niche modelling: applying joint hierarchical niche models to ecological networks. *Ecology Letters*, 28(6), e70120.

Lai, H. R., Hill, T., Stivanello, S., Chapman, H. M. (2025). Changes in quantity and timing of foliar and reproductive phenology of tropical dry-forest trees under a warming and drying climate. *Journal of Ecology*, 113(6), 1560–1572.

∗ Lai, H. R., Chong, K. Y., Wong, M. Q. Y., Tan, H. T. W., Yeo, D. C. J. (2025). Litter Depth, Not Canopy Openness, Mediates the Occurrence of an Invasive Shrub From the Forest Edge to Interior. *Biotropica*, 57, e70002.

∗ Lai, H. R., Bellingham, P. J., McCarthy, J. K., Richardson, S. J., Wiser, S. K., Stouffer, D. B. (2024). Detecting Nonadditive Biotic Interactions and Assessing Their Biological Relevance among Temperate Rainforest Trees. *American Naturalist*, 204(2), 105–120.

∗ O'Brien, S. A., Anderson, D. P., Lavorel, S., Lai, H. R., de Burgh, N., Tylianakis, J. M. (2024). Landscape patterns drive provision of nature's contributions to people by mobile species. *Journal of Applied Ecology*, 61, 2666–2678.

∗ Martins, L. P., Garcia-Callejas, D., Lai, H. R., Wootton, K. L., Tylianakis, J. M. (2024). The propagation of disturbances in ecological networks. *Trends in Ecology and Evolution*, 39(6), 558–570.

∗ Radford Smith, J., Lai, H. R., Cathcart-van Weeren, E., Dwyer, J. (2024). A Functional Basis for the Assembly of Australian Subtropical Rainforest Tree Communities. *Ecology Letters*, 27, e70014.

∗ Lai, H. R., Chong, K. Y., Yee, A.T.K. (2022). Ten years after: what we learned from the Mandai storm forest. *Nature in Singapore*, Supplement(e2022121), 207–218.

∗ Hall, J. S., Plisinski, J. S., Mladinich, S. K., van Breugel, M., Lai, H. R., Asner, G. P., Walker, K., Thompson, J. R. (2022). Deforestation scenarios show the importance of secondary forest for meeting Panama's carbon goals. *Landscape Ecology*, 37(3), 673–694.

Lam, W. N., Chan, P. J., Ting, Y. Y., Sim, H. J., Lian, J. J., Chong, R., Rahman, N. E., Tan, L. W. A., Ho, Q. Y., Chiam, Z., Arora, S., Lai, H. R., Ramchunder, S. J., Peh, K. S. H., Cai, Y., Chong, K. Y. (2022). Habitat Adaptation Mediates the Influence of Leaf Traits on Canopy Productivity: Evidence from a Tropical Freshwater Swamp Forest. *Ecosystems*, 25(5), 1006–1019.

Lai, H. R., Chong, K. Y., Yee, A. T. K., Mayfield, M. M., Stouffer, D. B. (2022). Non-additive biotic interactions improve predictions of tropical tree growth and impact community size structure. *Ecology*, 103(2), e03588.

∗ Lai, H. R., Tan, G. S. Y., Neo, L., Kee, C. Y., Yee, A. T. K., Tan, H. T. W., Chong, K. Y. (2021). Decoupled responses of native and exotic tree diversities to distance from old-growth forest and soil phosphorus in novel secondary forests. *Applied Vegetation Science*, 24(1), e12548.

Lai, H. R., Craven, D., Hall, J. S., Hui, F. K. C., van Breugel, M. (2021). Successional syndromes of saplings in tropical secondary forests emerge from environment-dependent trait–demography relationships. *Ecology Letters*, 24(9), 1776–1787.

∗ Falster, A. D., Gallagher, R., Wenk, E., Wright, I., Indiarto, D., ..., Lai, H. R., ... Kasia Ziemińska. (2021). AusTraits – a curated plant trait database for the Australian flora. *Nature Scientific Data*, 8(1), 254.

∗ Song, X. P., Lai, H. R., Wijedasa, L. S., Tan, P. Y., Edwards, P. J., Richards, D. R. (2020). Height-diameter allometry for the management of city trees in the tropics. *Environmental Research Letters*, 15(11), 114017.

Lai, H. R., Chong, K. Y., Yee, A. T. K., Tan, H. T. W., van Breugel, M. (2020). Functional traits that moderate tropical tree recruitment during post-windstorm secondary succession. *Journal of Ecology*, 108(4), 1322–1333.

∗ Yee, A. T. K., Lai, H. R., Chong, K. Y., Neo, L., Koh, C. Y., Tan, S. Y., Seah, W. W., Loh, J. W., Lim, R. C. J., van Breugel, M., Tan, H. T. W. (2019). Short-term responses in a secondary tropical forest after a severe windstorm event. *Journal of Vegetation Science*, 30(4), 720–731.

Chiam, Z., Song, X. P., Lai, H. R., Tan, H. T. W. (2019). Particulate matter mitigation via plants: Understanding complex relationships with leaf traits. *Science of the Total Environment*, 688, 398–408.

∗ van Breugel, M., Craven, D., Lai, H. R., Baillon, M., Turner, B. L., Hall, J. S. (2019). Soil nutrients and dispersal limitation shape compositional variation in secondary tropical forests across multiple scales. *Journal of Ecology*, 107(2), 566–581.

Wainwright, C. E., HilleRisLambers, J., Lai, H. R., Loy, X., Mayfield, M. M. (2019). Distinct responses of niche and fitness differences to water availability underlie variable coexistence outcomes in semi-arid annual plant communities. *Journal of Ecology*, 107(1), 293–306.

Lam, W. N., Lai, H. R., Lee, C. C., Tan, H. T. W. (2018). Evidence for pitcher trait-mediated coexistence between sympatric Nepenthes pitcher plant species across geographical scales. *Plant Ecology and Diversity*, 11(3), 283–294.

Bimler, M. D., Stouffer, D. B., Lai, H. R., Mayfield, M. M. (2018). Accurate predictions of coexistence in natural systems require the inclusion of facilitative interactions and environmental dependency. *Journal of Ecology*, 106(5), 1839–1852.

Lai, H. R., Hall, J. S., Batterman, S. A., Turner, B. L., van Breugel, M. (2018). Nitrogen fixer abundance has no effect on biomass recovery during tropical secondary forest succession. *Journal of Ecology*, 106(4), 1415–1427.

Wainwright, C. E., Staples, T. L., Charles, L. S., Flanagan, T. C., Lai, H. R., Loy, X., Reynolds, V. A., Mayfield, M. M. (2018). Links between community ecology theory and ecological restoration are on the rise. *Journal of Applied Ecology*, 55(2), 570–581.

∗ Sams, M. A., Lai, H. R., Bonser, S. P., Vesk, P. A., Kooyman, R. M., Metcalfe, D. J., Morgan, J. W., Mayfield, M. M. (2017). Landscape context explains changes in the functional diversity of regenerating forests better than climate or species richness. *Global Ecology and Biogeography*, 26(10), 1165–1176.

Lai, H. R., Hall, J. S., Turner, B. L., van Breugel, M. (2017). Liana effects on biomass dynamics strengthen during secondary forest succession. *Ecology*, 98(4), 1062–1070.

Lai, H. R., Mayfield, M. M., Gay-des-combes, J. M., Spiegelberger, T., Dwyer, J. M. (2015). Distinct invasion strategies operating within a natural annual plant system. *Ecology Letters*, 18(4), 336–346.

### *Software*

Lai, H. R.. (2025) `SGtraits`: A trait database for the Singapore flora (v0.0.2). *Zenodo*.

Virtual Ecosystem Project Team. (2024) `Virtual Ecosystem`.

Lai, H. R., Chong K. Y., Yee A. T. K. (2020) `novelforestSG`: Data and model for Lai, Chong et al. (2020) *Appl. Veg. Sci.* R package version 2.0.

Song, X. P., Lai, H. R. (2020) `allometree`: Allometric Scaling of Urban Trees. R package version 0.1.

## Conferences & Presentations

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| 2025 | Invited talks. National University of Singapore and National Parks Board Singapore. |
| 2024 | Ecological Society of Australia. Melbourne, Australia. |
| 2022 | sUCCESS secondary forest workshop. Leipzig, Germany. |
| 2022 | Association of Tropical Biology and Conservation (ATBC). Cartagena, Colombia. |
| 2020 | International Statistical Ecology Conference (ISEC). virtual. |
| 2018 | Association of Tropical Biology and Conservation (ATBC). Kuching, Malaysia. |
| 2016 | Conservation Asia. Singapore. |
| 2015 | 20th Biological Science Graduate Congress. Bangkok, Thailand. |

## Grants, Awards & Scholarships

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| 2026–2027 | Forest Global Earth Observatory Research Award (USD 10,000), The Smithsonian Institute, USA |
| 2022–2025 | Marsden Fast Start Fund (NZD 360,000), Royal Society Te Apārangi, New Zealand |
| 2019 | International Society for Plant Molecular Biology Medal for most outstanding thesis |
| 2019 | CUGE Visiting Research Fellowship, National Parks Board Singapore |
| 2017 & 2018 | Teaching Assistant Award, National University of Singapore |
| 2016 | City Developments Limited (CDL) Urban Ecology and Conservation Scholarship |
| 2013 | University Medal, University of Queensland |
| 2011 | Summer Research Scholarship, University of Queensland |
| 2010 | D.A. Herbert Prize in Botany & F.A. Perkins Prize in Entomology, University of Queensland |

## Services & Outreach

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| Membership | Association of Tropical Biology and Conservation (ATBC), Open Traits Network (OTN), Ecological Forecast Initiative (EFI) |
| Peer review | *Science*, *Journal of Ecology*, *Functional Ecology*, *Ecology*, *Methods in Ecology and Evolution*, *Ecological Applications*, *Ecography*, *Biodiversity and Conservation*, *Biological Invasions*, *Science of the Total Environment*, *Journal of Vegetation Science*, *Global Ecology and Biogeography*, *Global Change Biology* etc. |
| 2020 | `RMarkdown` tutorial, University of Canterbury, New Zealand |
| 2017 | Science Research Programme, Hwa Chong Institution, Singapore |
| 2016 | Festival of Biodiversity, Singapore Botanic Gardens |

## Languages

English, Bahasa Malaysia, Mandarin, Cantonese

## Other skills

Geographic information system (with QGIS and R), spatial modelling, Bayesian inference, version control (GitHub), plant survey and identification

## Professional references

Robert Ewers (current post-doc supervisor)
Professor, Imperial College London, United Kingdom
South East Asia Rainforest Research Partnership (SEARRP), Malaysia
Email: `r.ewers@imperial.ac.uk`

Jason Tylianakis (former post-doc supervisor & ongoing collaborator)
Professor, University of Canterbury, New Zealand
Email: `jason.tylianakis@canterbury.ac.nz`

Michiel van Bruegel (former PhD supervisor & ongoing collaborator)
Associate Professor, National University of Singapore, Singapore
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Alison Kim Shan Wee
Associate Professor, University of Nottingham Malaysia
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Nik Fadzly bin N Rosely
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